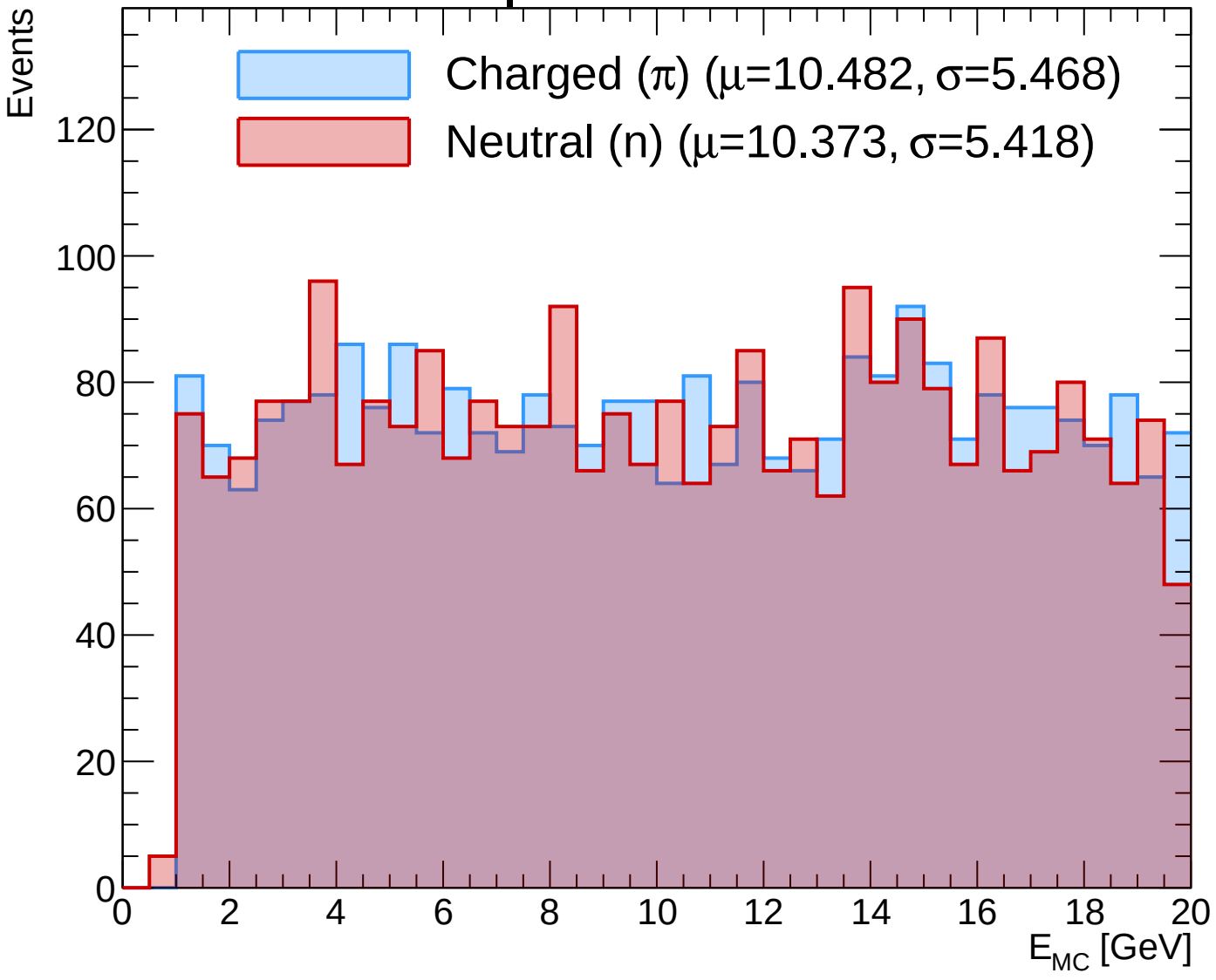
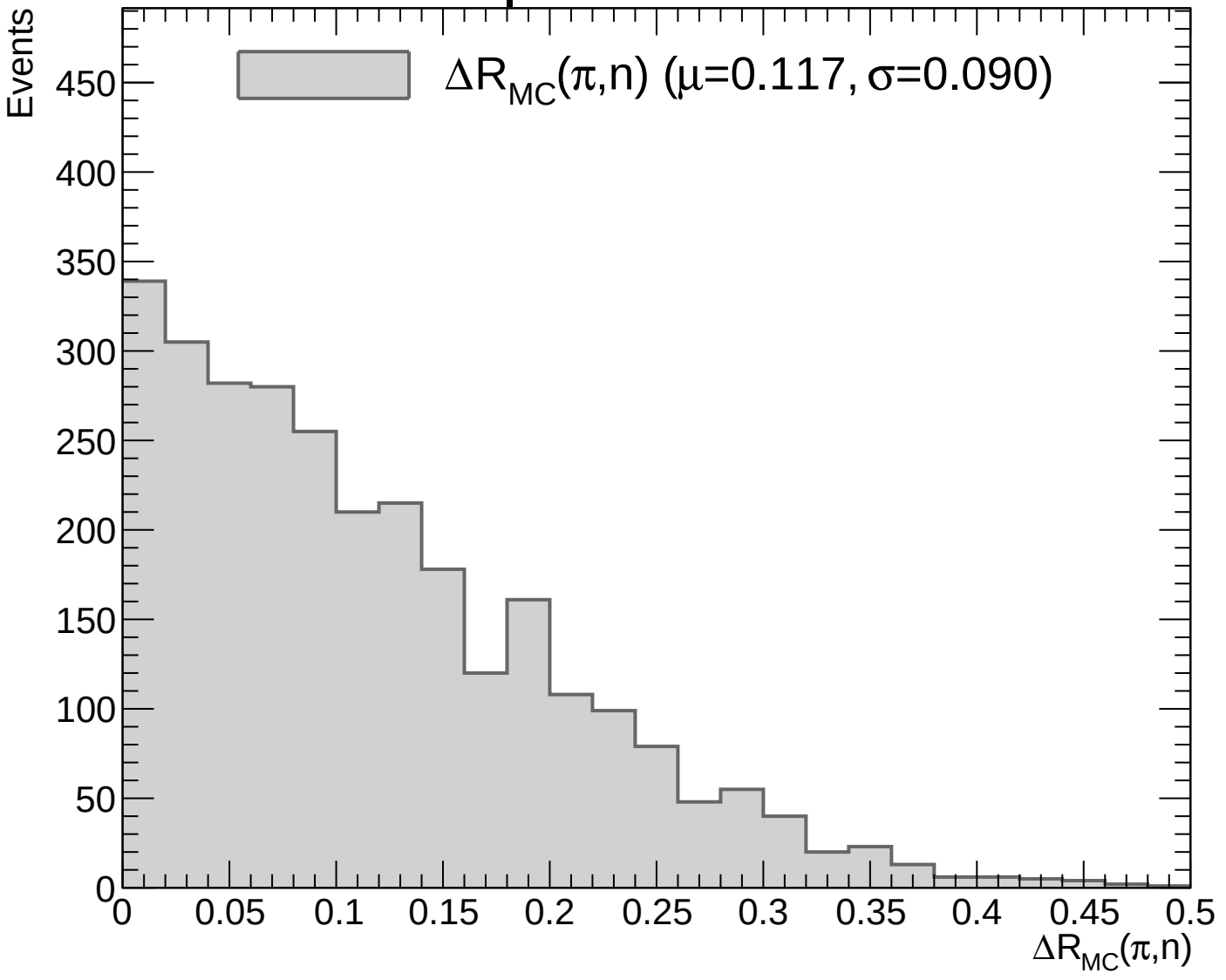
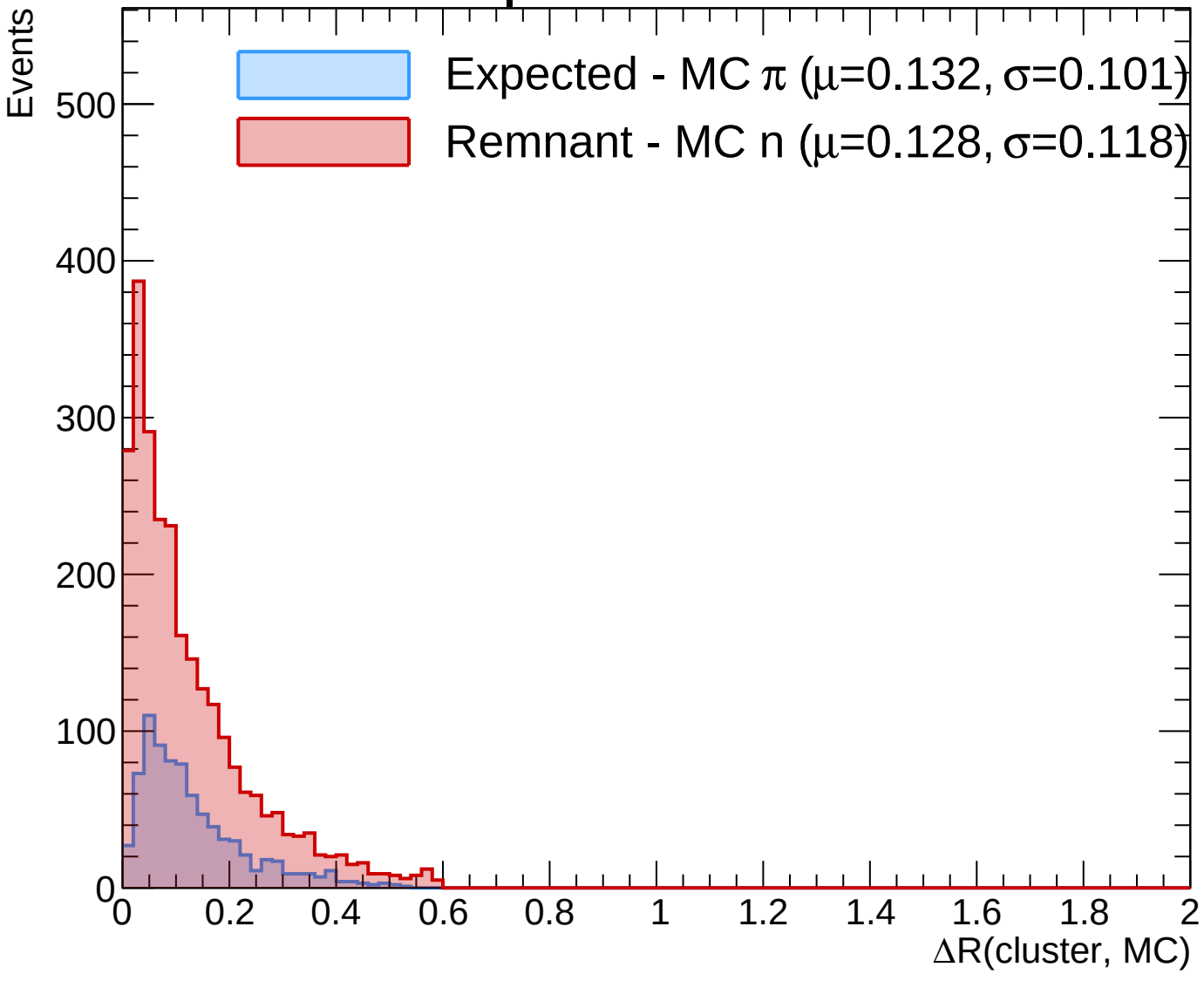
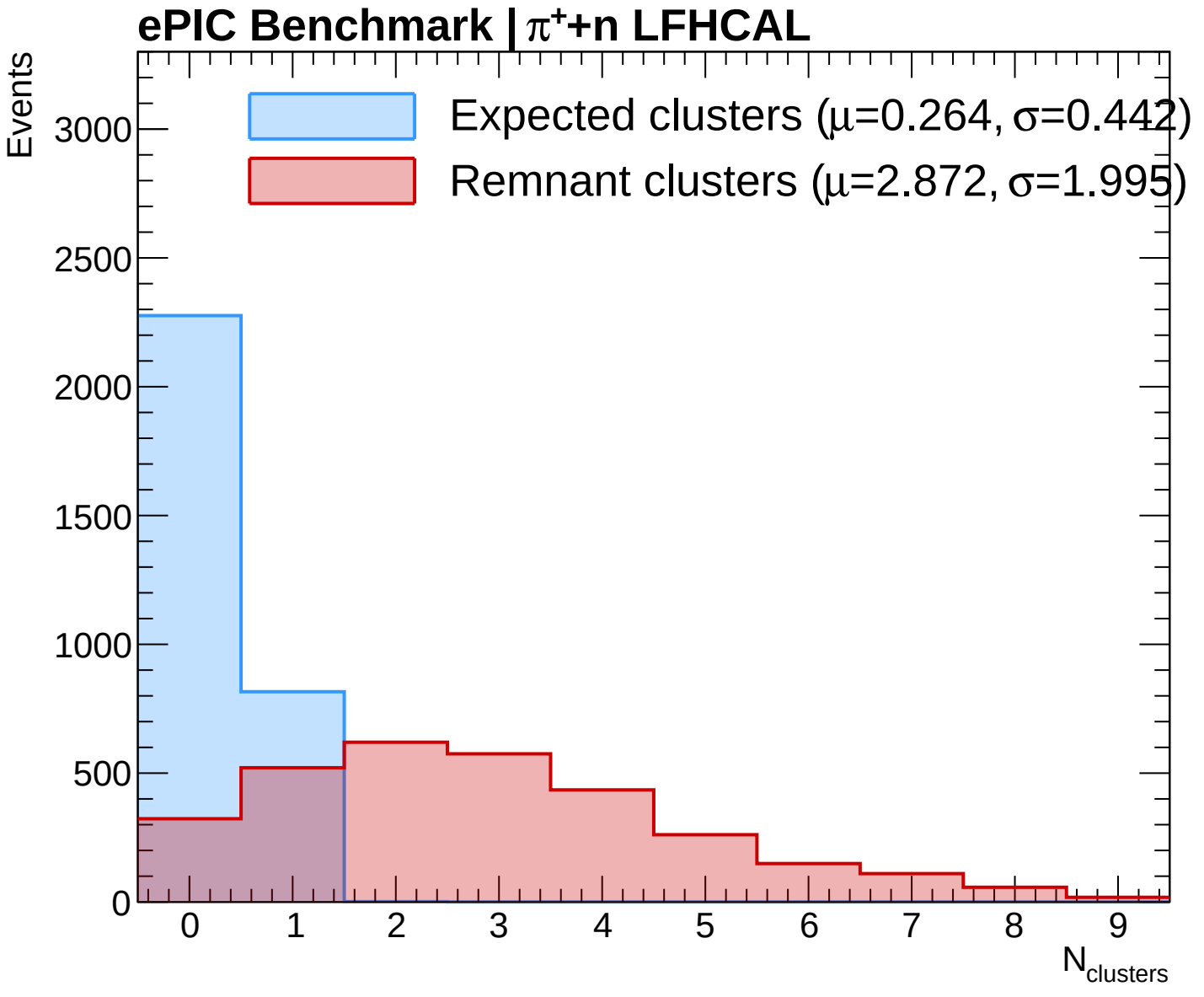
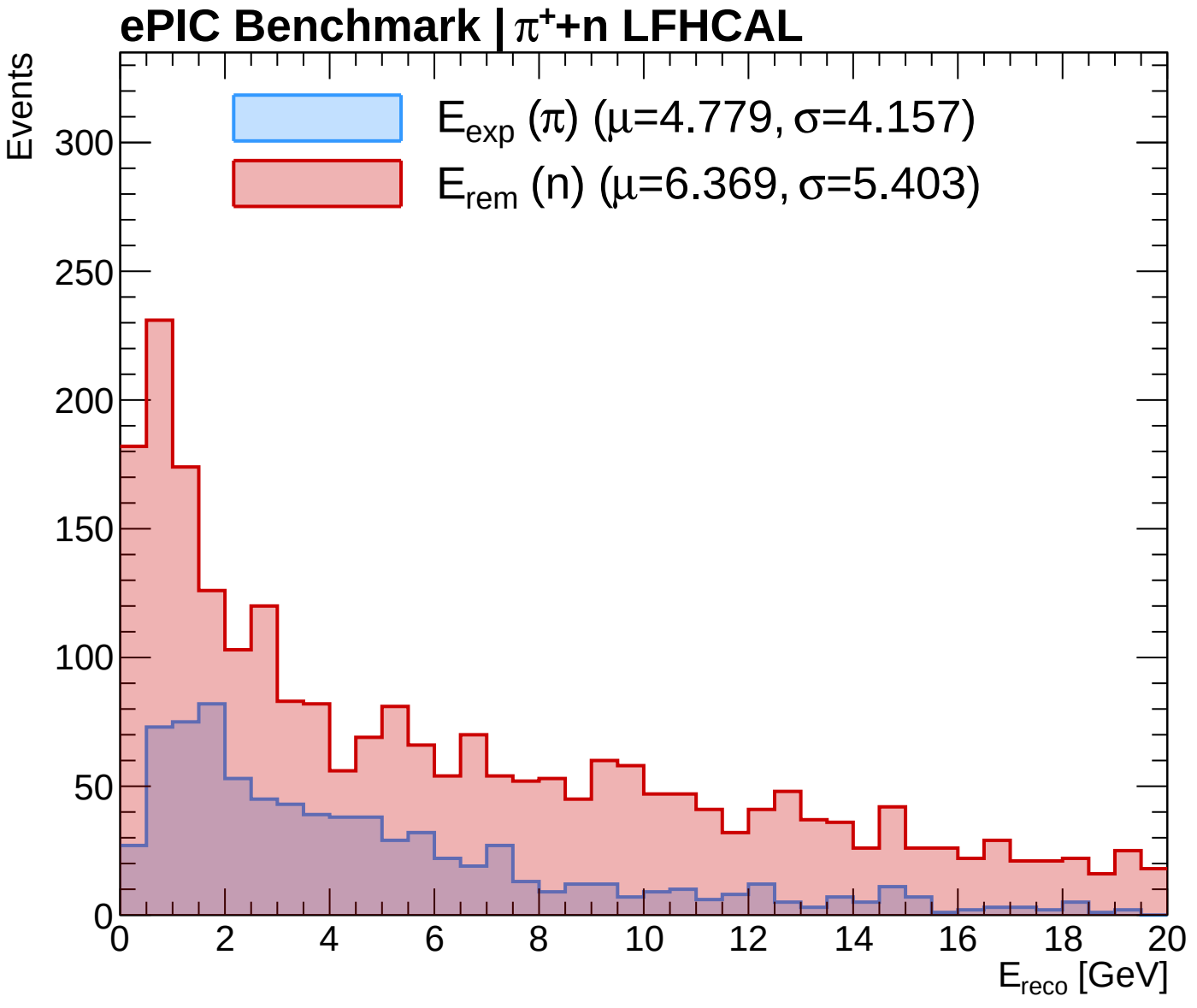
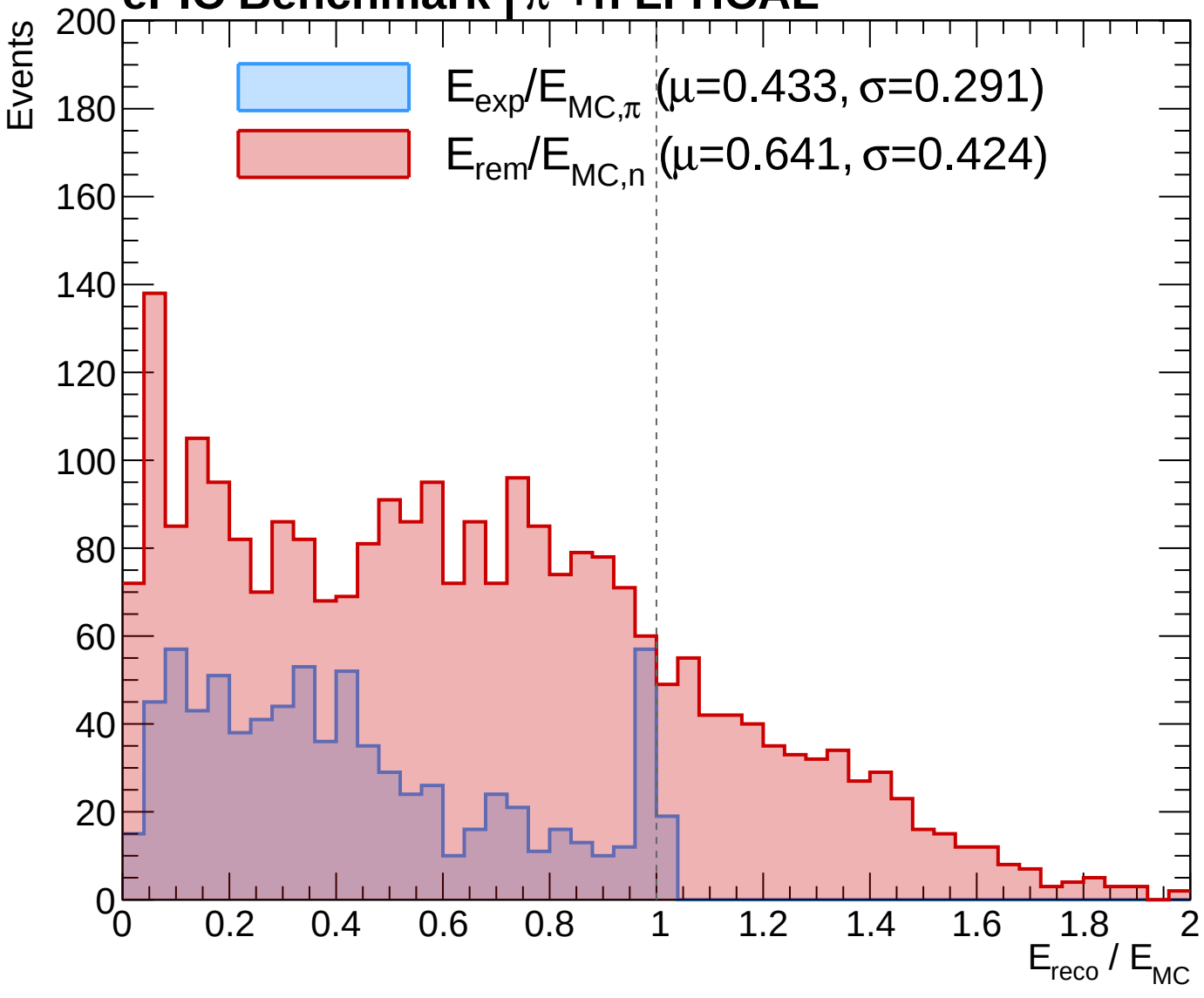

ePIC Benchmark | π⁺+n LFHCAL
Charged (π) (μ=10.482, σ=5.468)
Neutral (n) (μ=10.373, σ=5.418)
Events
0
20
40
60
80
100
120
0
2
4
6
8
10
12
14
16
18
20
E_MC [GeV]

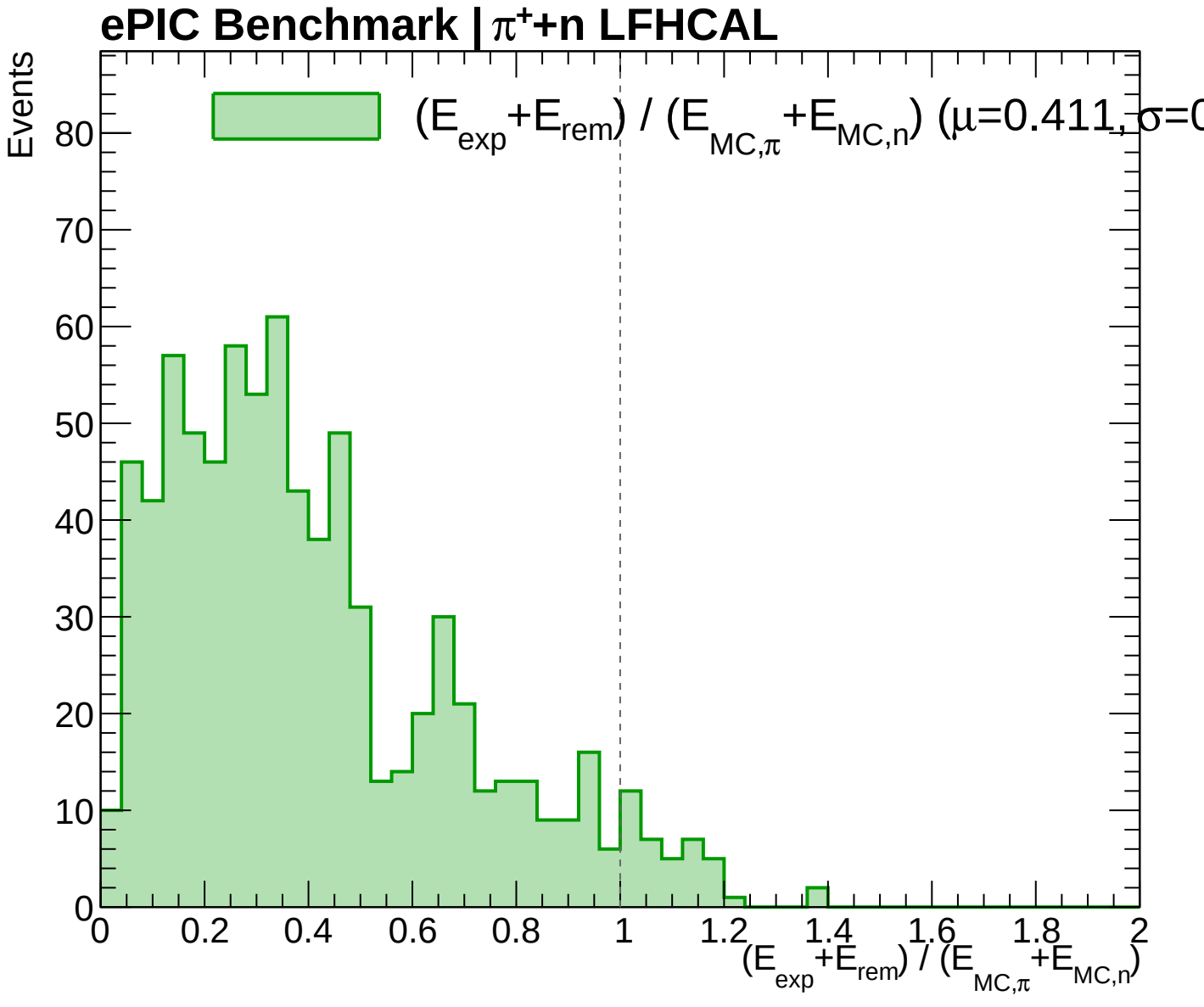

ePIC Benchmark | π⁺+n LFHCAL
ΔR_MC(π,n) (μ=0.117, σ=0.090)
Events
0
50
100
150
200
250
300
350
400
450
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(π,n)

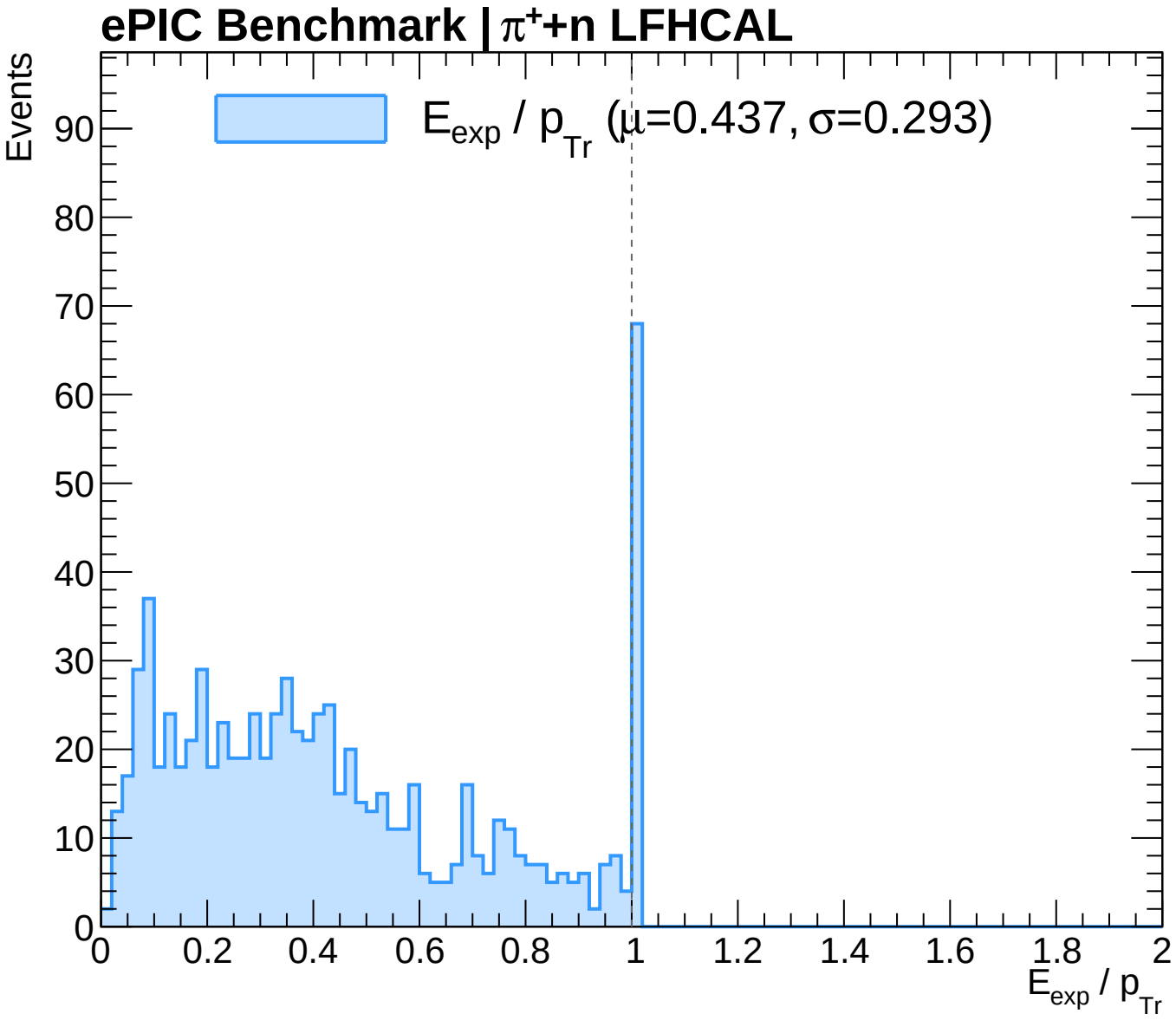

ePIC Benchmark | π⁺+n LFHCAL
Events
Expected - MC π (μ=0.132, σ=0.101)
Remnant - MC n (μ=0.128, σ=0.118)
0
100
200
300
400
500
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
ΔR(cluster, MC)

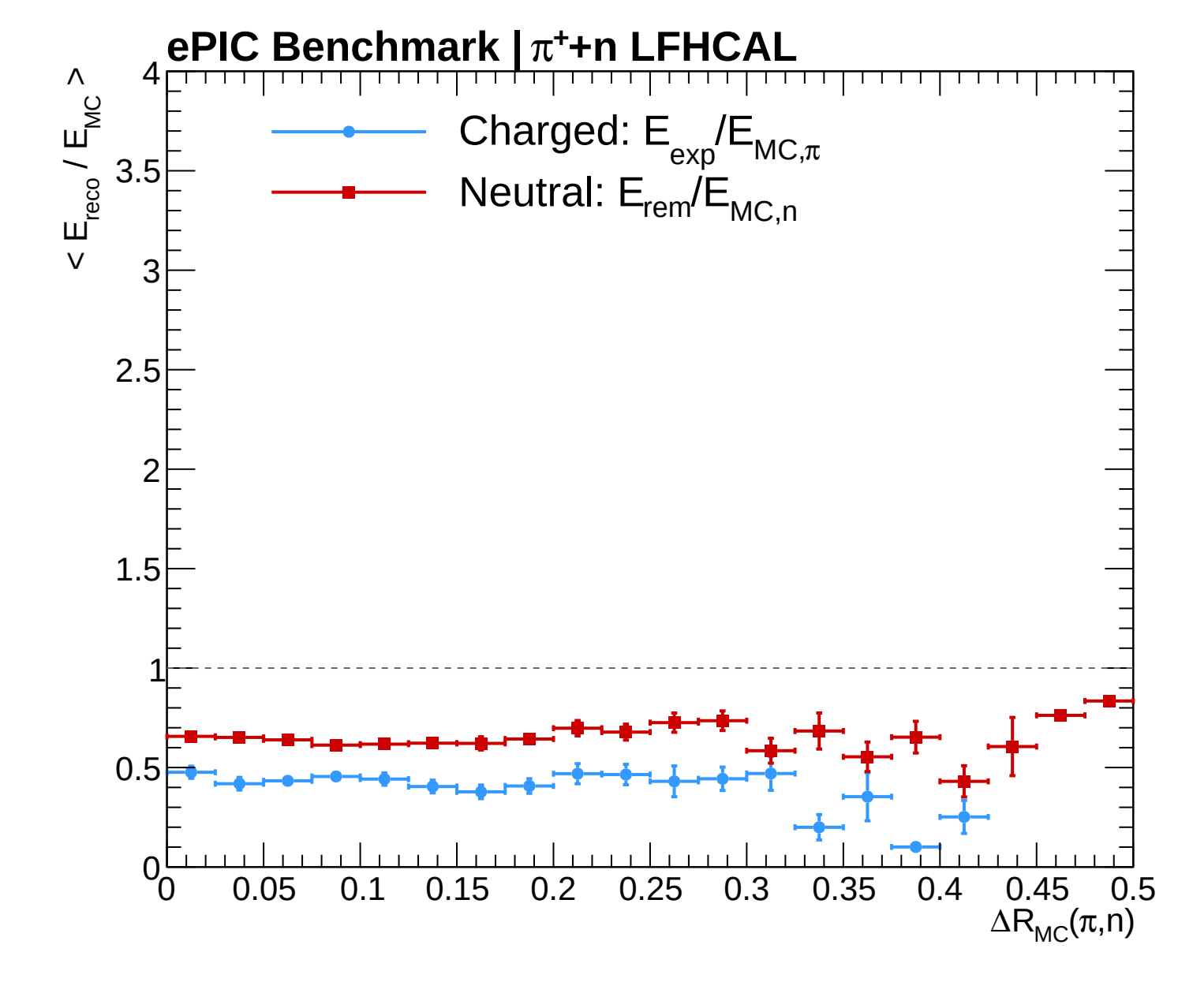

ePIC Benchmark | π⁺+n LFHCAL
Events
Expected clusters (μ=0.264, σ=0.442)
Remnant clusters (μ=2.872, σ=1.995)
3000
2500
2000
1500
1000
500
0
0
1
2
3
4
5
6
7
8
9
N clusters

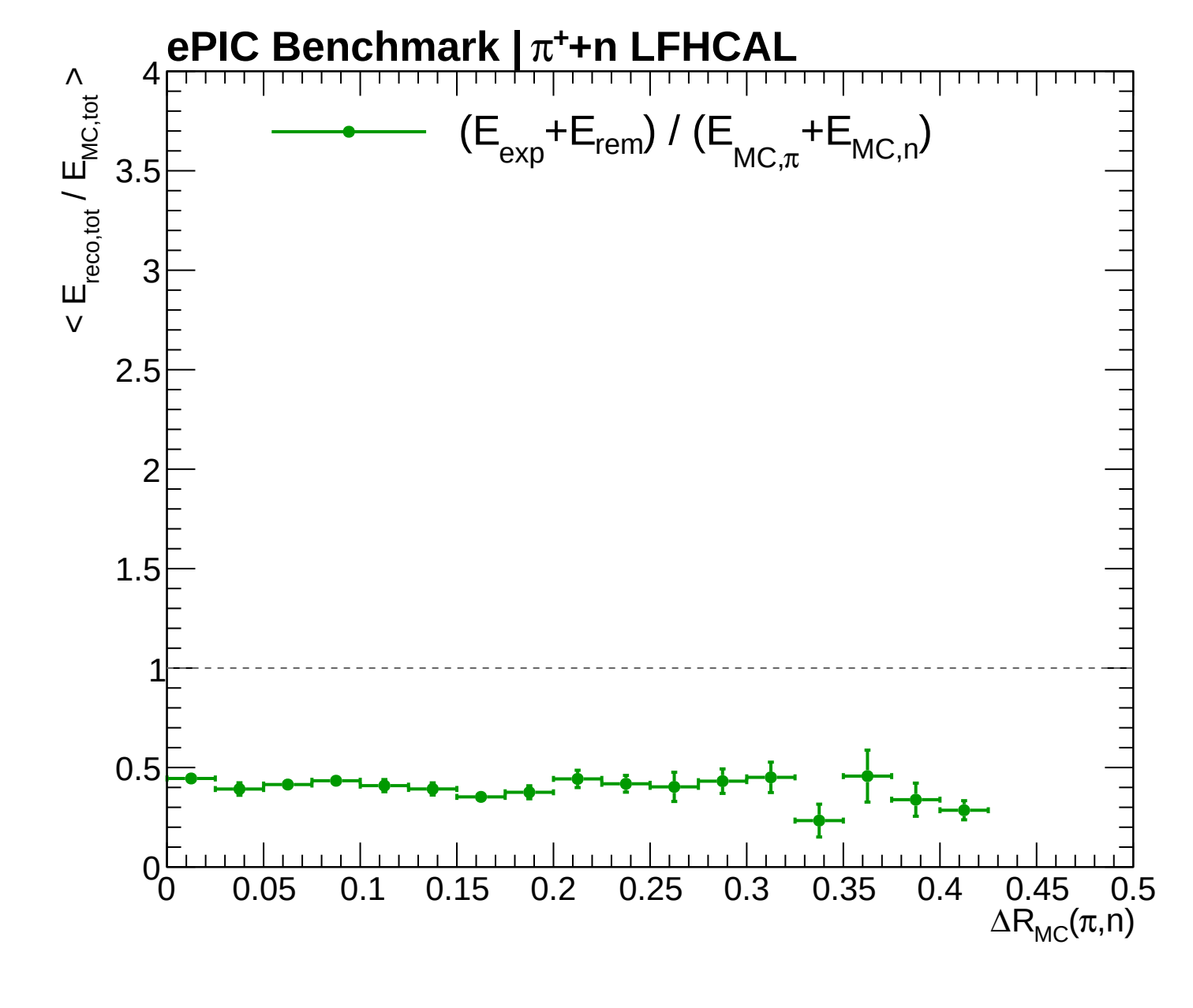

ePIC Benchmark | π⁺+n LFHCAL
Events
E_exp (π) (μ=4.779, σ=4.157)
E_rem (n) (μ=6.369, σ=5.403)
0
50
100
150
200
250
300
0
2
4
6
8
10
12
14
16
18
20
E_reco [GeV]

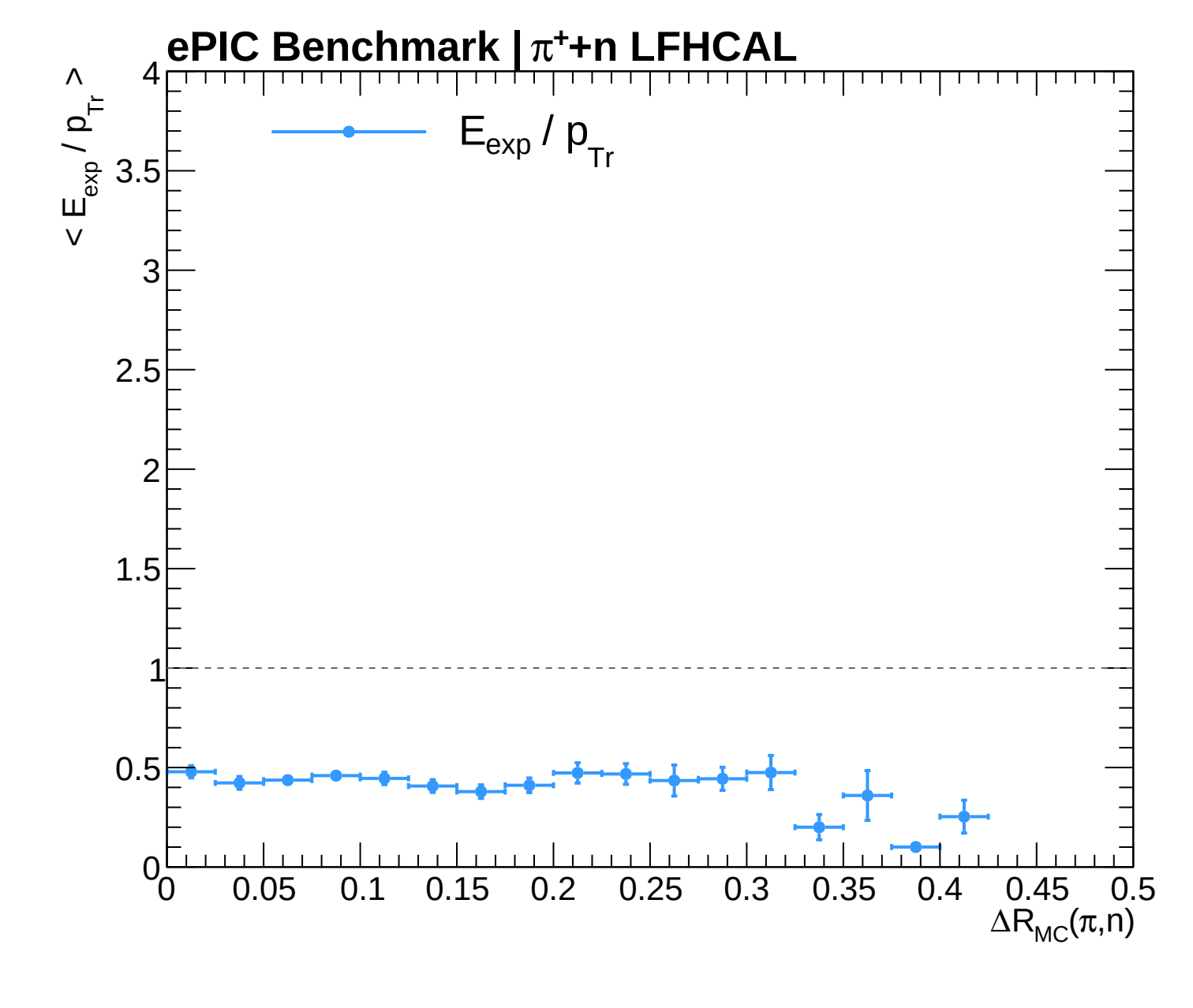

ePIC Benchmark | π⁺+n LFHCAL
Events
E_exp/E_MC,π (μ=0.433, σ=0.291)
E_rem/E_MC,n (μ=0.641, σ=0.424)
0
20
40
60
80
100
120
140
160
180
200
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_reco / E_MC

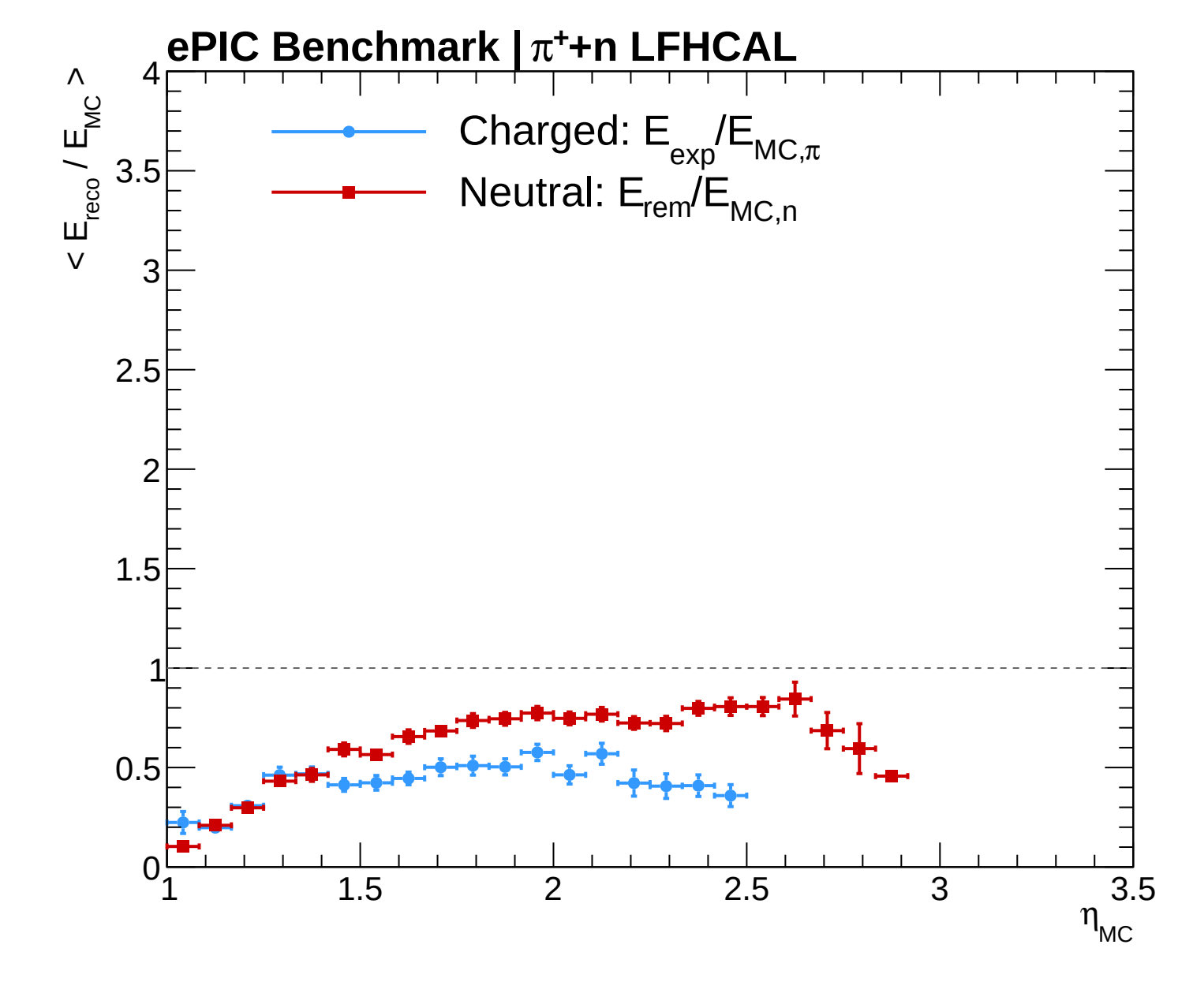

ePIC Benchmark | π⁺+n LFHCAL
Events
(E_exp+E_rem) / (E_MC,π+E_MC,n) (μ=0.411, σ=0
0
10
20
30
40
50
60
70
80
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
(E_exp+E_rem) / (E_MC,π+E_MC,n)

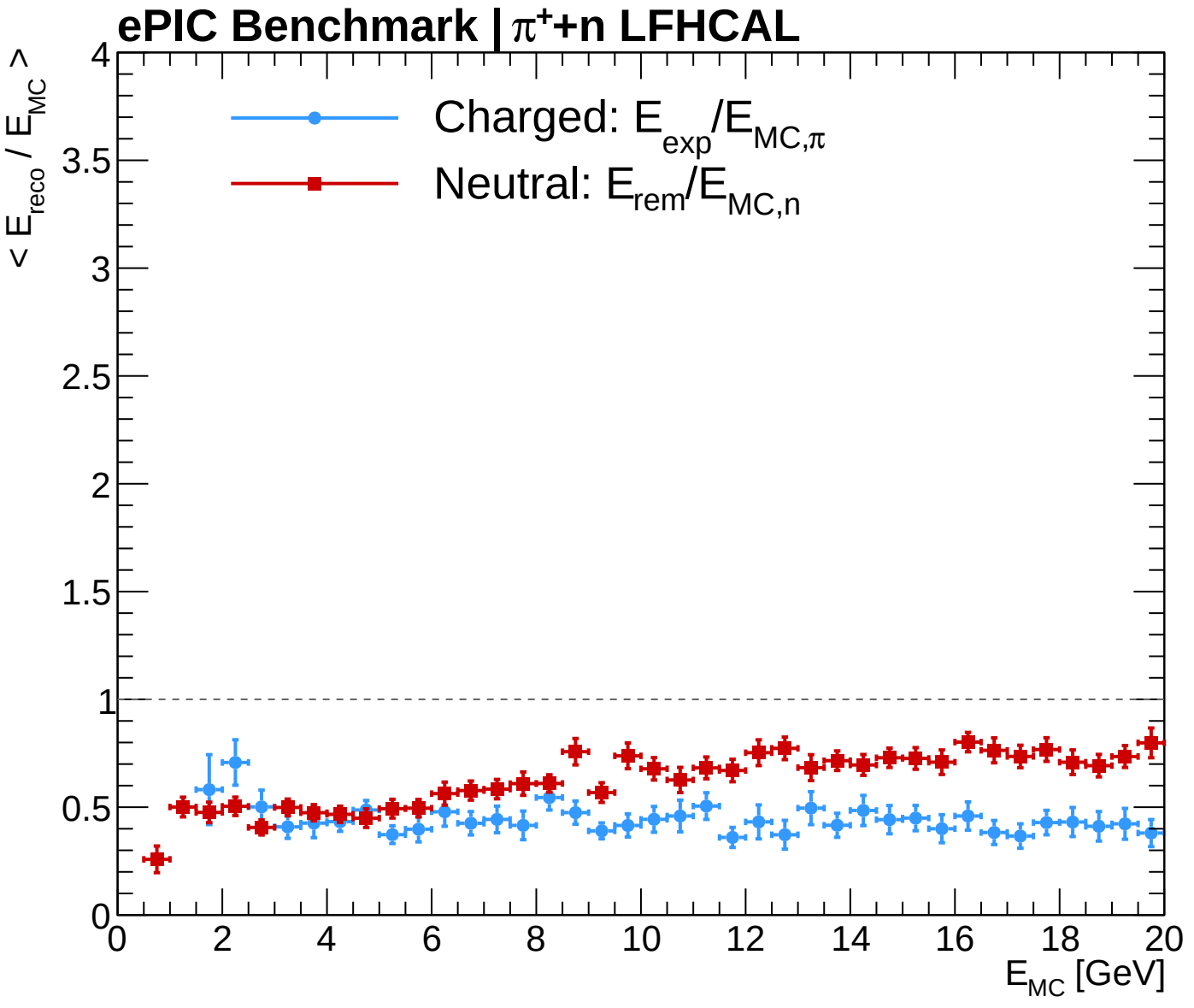

ePIC Benchmark | π⁺+n LFHCAL
Events
E_exp / p_Tr (μ=0.437, σ=0.293)
0
10
20
30
40
50
60
70
80
90
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / p_Tr

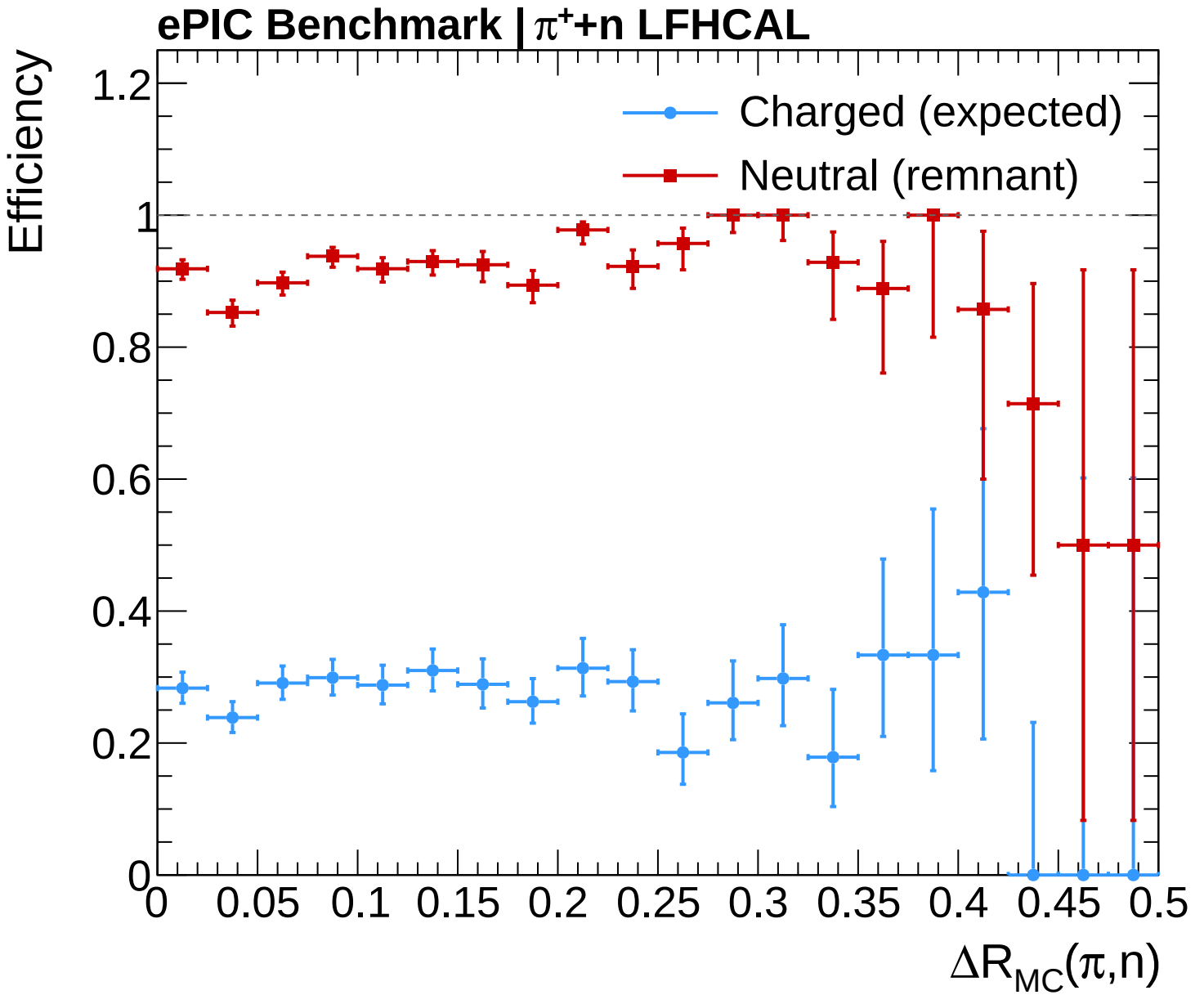

ePIC Benchmark | π⁺+n LFHCAL
Charged: E_exp/E_MC,π
Neutral: E_rem/E_MC,n
< E_reco / E_MC >
ΔR_MC(π,n)
0
0.5
1
1.5
2
2.5
3
3.5
4
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

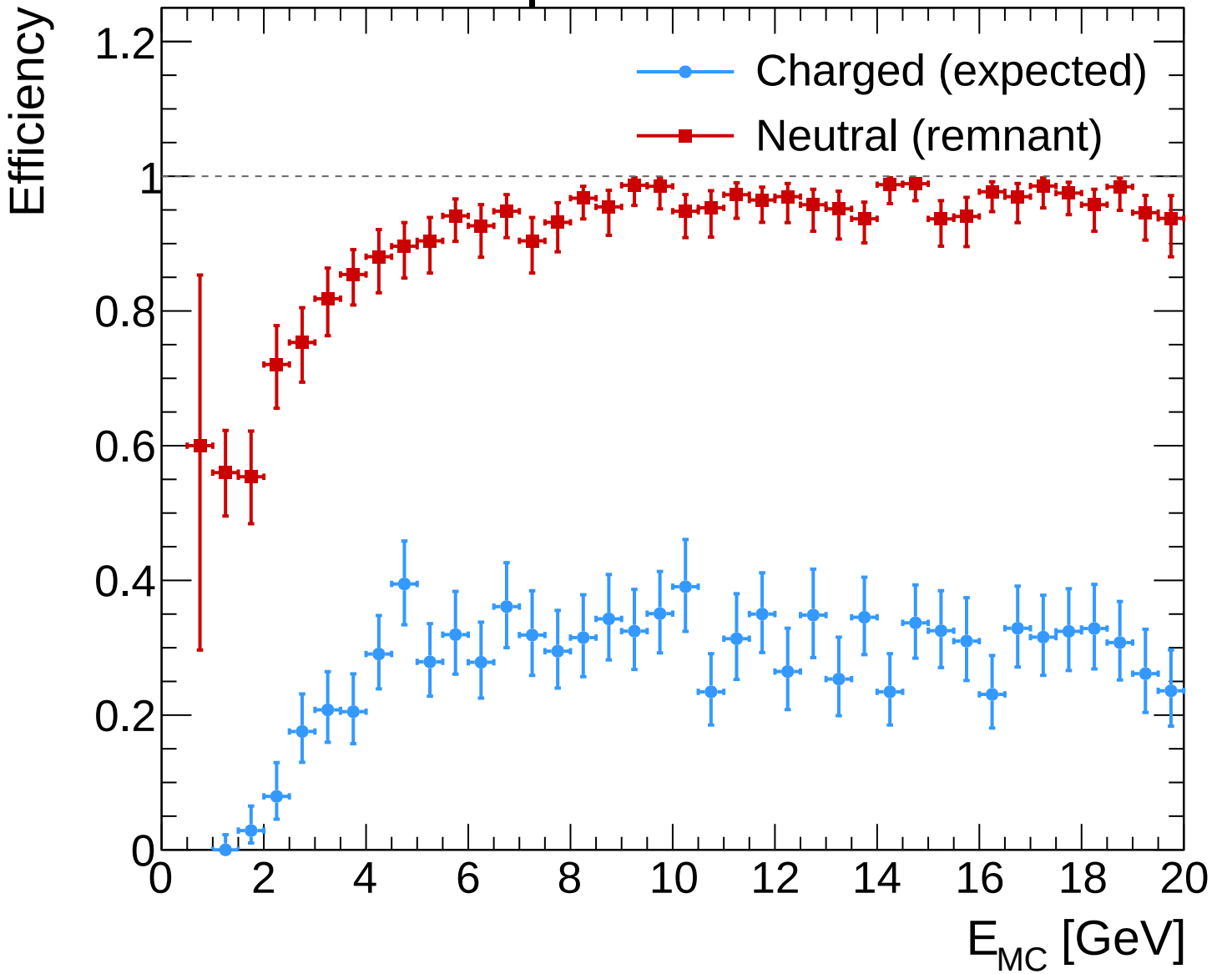

ePIC Benchmark | π⁺+n LFHCAL
(E_exp+E_rem) / (E_MC,π+E_MC,n)
< E_reco,tot / E_MC,tot >
ΔR_MC(π,n)
0
0.5
1
1.5
2
2.5
3
3.5
4
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

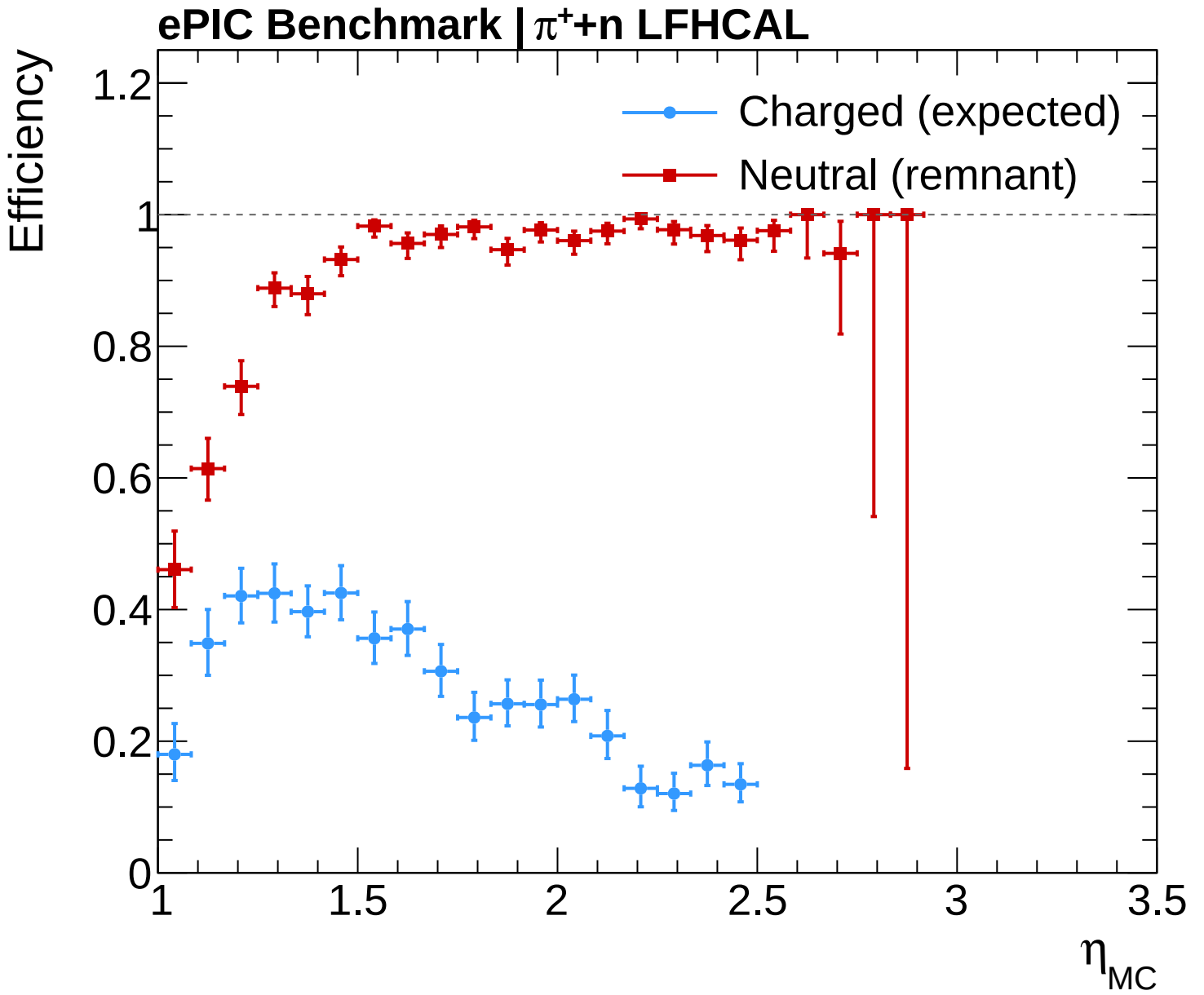

ePIC Benchmark | π⁺+n LFHCAL
< E_exp / p_Tr >
E_exp / p_Tr
4
3.5
3
2.5
2
1.5
1
0.5
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(π,n)


ePIC Benchmark | π⁺+n LFHCAL
Charged: E_exp/E_MC,π
Neutral: E_rem/E_MC,n
< E_reco / E_MC >
η_MC
0
0.5
1
1.5
2
2.5
3
3.5
4
1
1.5
2
2.5
3
3.5


ePIC Benchmark | π⁺+n LFHCAL
Charged: E_exp/E_MC,π
Neutral: E_rem/E_MC,n
< E_reco / E_MC >
E_MC [GeV]
0
0.5
1
1.5
2
2.5
3
3.5
4
0
2
4
6
8
10
12
14
16
18
20


ePIC Benchmark | π⁺+n LFHCAL
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(π,n)


ePIC Benchmark | π⁺+n LFHCAL
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
16
18
20
E_MC [GeV]


ePIC Benchmark | π⁺+n LFHCAL
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
1
1.5
2
2.5
3
3.5
η_MC